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DEPARTMENT OF COMPUTER ENGINEERING

Mini Project

on

"Smart Used Car Price Estimation System"

- Submitted by -

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1. Title

Smart Used Car Price Estimation System

A machine learning project that studies details of second-hand cars, predicts a fair selling price, groups cars into segments, finds similar cars and flags unusual listings.

2. Abstract

Buying or selling a second-hand car is difficult because most people do not know a fair price. Buyers may pay too much and sellers may sell too cheap. Prices depend on many things together, such as the age of the car, the kilometres it has run, the fuel type, the transmission, the number of previous owners and the brand. This project builds a simple system that estimates a fair price using machine learning.

First, the data is cleaned and new features are created, such as car age, kilometres per year and a brand value. Then five models are trained to predict the price: Multiple Linear Regression, Polynomial Regression, a single Decision Tree, k-Nearest Neighbors Regression and AdaBoost Regressor. The models are compared using R^2 score, Mean Absolute Error and Root Mean Squared Error. K-Means clustering groups the cars into Budget, Mid-range and Premium segments, and DBSCAN flags unusual listings that do not match similar cars.

For any new car, the system gives a predicted price, its segment and the three most similar cars from the data. The results, including graphs and a comparison table of the models, are given in the Results section.

Keywords: Used car, price prediction, Multiple Linear Regression, Polynomial Regression, k-NN Regression, AdaBoost, K-Means, DBSCAN.

3. Introduction

Machine learning helps computers learn patterns from past data and use them to make predictions on new data. Used car prices are a good example. A car loses value every year, loses more value when it has run many kilometres, and is worth more when it is from a trusted brand or has fewer previous owners. A computer can learn these patterns from thousands of past sales.

In this project we use four ideas from our Machine Learning Laboratory (CS3201L):

- **Multiple Linear Regression and Polynomial Regression (Assignment 2):** predict a number from many inputs. The polynomial model can follow a curve, such as the way price falls quickly in the first few years.
- **k-Nearest Neighbors (Assignment 5):** predicts the price from the most similar cars. We also study how the value of K changes the result.
- **AdaBoost (Assignment 9):** combines many simple decision trees into a stronger model and is compared with a single Decision Tree.
- **K-Means (Assignment 7) and DBSCAN (Assignment 8):** K-Means groups cars into segments. DBSCAN finds listings that do not fit any group.

Together these methods make a complete workflow: data cleaning, feature engineering, model building, evaluation and prediction. This is the workflow expected in the mini project (Assignment 10).

4. Problem Statement

To study the details of second-hand cars, estimate a fair selling price, show similar cars that support the price, group cars into Budget, Mid-range and Premium segments, and flag unusual listings, so that buyers, sellers and dealers can take better decisions.

Who can use it?

- Students and families: to check if the price of a car or bike is fair before buying.
- Sellers: to decide a realistic asking price.
- Car dealers: to buy and sell at the right price and spot overpriced or underpriced listings.
- Teachers and learners: as a simple example of regression, clustering and boosting.

5. Objectives

- To clean the car data by removing duplicates, filling missing values, converting owner text to numbers and removing extreme outliers.
- To create useful features such as car age, kilometres per year and brand value.
- To predict the price using Multiple Linear Regression, Polynomial Regression, Decision Tree, k-NN Regression and AdaBoost Regressor.
- To choose the best value of K for k-NN using cross-validation, and the best settings for AdaBoost.
- To compare all models using R^2 score, MAE and RMSE.
- To group cars into Budget, Mid-range and Premium segments using K-Means.
- To flag unusual listings using DBSCAN.
- To predict the price of a new car, show similar cars, and visualise the results with graphs.

6. Data Set and Libraries Required

6.1 Data set

The project uses a table of used cars with one row per car, for example the CarDekho vehicle dataset from Kaggle saved as a CSV file named **cars.csv**. Any file with the columns below can be used; only the column names at the top of the code need to match.

Column	Meaning	Type
name	Brand and model of the car (the first word is taken as the brand)	Text
year	Year in which the car was bought	Number
selling_price	Price at which the car was sold (rupees)	Number
km_driven	Total kilometres the car has run	Number
fuel	Petrol, Diesel, CNG, LPG and so on	Text
seller_type	Individual or Dealer	Text
transmission	Manual or Automatic	Text
owner	First Owner, Second Owner and so on	Text

Features created by us: **car_age** (current year minus year), **km_per_year** (km_driven divided by car_age), **owner_num** (owner as a number) and **brand_value** (median price of that brand in the training data).

6.2 Libraries required

Library	Used for
Python 3.x	Programming language
NumPy	Numerical calculations
Pandas	Reading the data, cleaning, creating features
Matplotlib	Drawing graphs
scikit-learn	Scaling, train-test split, Multiple Linear and Polynomial Regression, Decision Tree, k-NN, AdaBoost, K-Means, DBSCAN, cross-validation and evaluation metrics

Install command: `pip install numpy pandas matplotlib scikit-learn`

7. Approach / Methodology

The project follows eight steps. Each step feeds the next one.

Step	What we do	Method used
1	Load the data, remove duplicates, drop rows with missing name, year or price, fill missing kilometres with the median, convert owner text (First Owner, Second Owner) to numbers, remove the 1% cheapest and 1% costliest cars and extreme kilometre values	Pandas
2	Create car_age, km_per_year and brand (first word of the name)	Feature engineering
3	Split the data 80% for training and 20% for testing. Calculate brand_value from the training data only. Convert fuel, seller type and transmission to 0/1 columns. Scale all features to the same range	train_test_split, get_dummies, StandardScaler
4	Train the models on the **logarithm of the price**: Multiple Linear Regression, Polynomial Regression (degree 2), Decision Tree, k-NN Regression (K chosen by 5-fold cross-validation) and AdaBoost Regressor (settings chosen by grid search)	Assignments 2, 5, 9
5	Convert predictions back to lakh rupees and evaluate every model on the test data using R^2, MAE and RMSE	scikit-learn metrics
6	Group cars into three clusters using price, age and kilometres. Name them Budget, Mid-range and Premium by their average price	K-Means (Assignment 7)
7	Find listings that do not belong to any dense group of similar cars	DBSCAN (Assignment 8)
8	For a new car: predict the price with the best model, find its segment, show the 3 most similar cars and draw the graphs	NearestNeighbors, Matplotlib

7.1 Why do we use the logarithm of the price?

A few cars are very expensive and most cars are cheap. If we use the price directly, the model pays too much attention to the costly cars. Taking the logarithm makes the price range smaller and fairer, so a mistake of 10% is treated alike for a cheap car and for a costly car. After prediction we convert the answer back to normal rupees.

7.2 How each algorithm is used

Algorithm	Job in this project
Multiple Linear Regression	Baseline. Price is a weighted sum of the features.
Polynomial Regression	Captures curves, for example price falling fast in the first years and slowly later.
Decision Tree	A single tree. Used as the comparison model for AdaBoost, as asked in Assignment 9.
k-NN Regression	Predicts the average price of the K most similar cars. K is chosen by cross-validation.
AdaBoost Regressor	Builds many small trees one after another. Each new tree focuses on the cars that earlier trees predicted badly.
K-Means	Groups cars into Budget, Mid-range and Premium.
DBSCAN	Marks listings that are far from all dense groups as unusual.

7.3 Evaluation measures in simple words

Measure	Meaning
R^2 score	How much of the price changes the model explains. Closer to 1 is better.
MAE (Mean Absolute Error)	Average size of the mistake in lakh rupees. Smaller is better.
RMSE (Root Mean Squared Error)	Like MAE but punishes big mistakes more. Smaller is better.

10. Conclusion

In this mini project we built a complete machine learning workflow for used car prices. We cleaned the data, created features such as car age and brand value, and trained five models to predict the price. The models were compared using R^2 score, MAE and RMSE, so the best model for this data could be chosen fairly. K-Means grouped the cars into Budget, Mid-range and Premium segments, and DBSCAN flagged unusual listings.

The project also shows that the best model depends on the data. Simple models are easy to understand, curved models can follow the fall in price with age, and boosting can improve accuracy by learning from earlier mistakes. For a new car, the system gives a predicted price, a segment and similar cars, which helps a buyer or seller to trust the number.

Future scope:

- Use the full real dataset with more details such as mileage, engine size and number of seats.
- Try other boosting methods such as Gradient Boosting and tune the model settings further.
- Build a small web or mobile app where a user enters car details and gets the price.
- Update the model regularly with new listings so that it follows market prices.

References

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